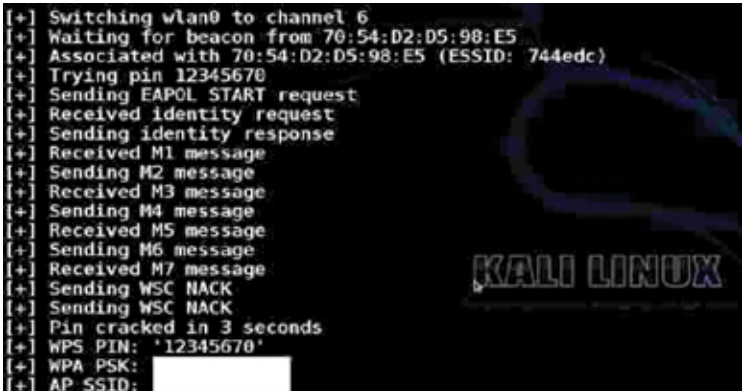


- Fool a user into giving you the password

There are smarter ways of getting the password directly from the user like the Man in the Middle attack. It involves phishing. In this clever attack, you forcefully disconnect a client from the original WPA-2 network, then trick them into connecting to a fake open network created by you. Once connected to the fake network, send the client a made up login page asking him to enter sensitive information. We cannot create a WPA-2 connection directly instead of an open network to get the password directly as WPA-2 performs a mutual authentication during the 4-way handshake. The client verifies whether the AP is legitimate or not, and knows the password (which is hashed), and the AP verifies that the client is legitimate and also knows the password. We don't have enough information to complete the 4-way handshake.

Drawback of hacking WPA/WPA-2

A WEP network could be hacked by any random guy with a laptop, and a network connection, hence the much more robust WPA/WPA2 was developed and adopted.



```
[+] Switching wlan0 to channel 6
[+] Waiting for beacon from 70:54:D2:D5:98:E5
[+] Associated with 70:54:D2:D5:98:E5 (ESSID: 744edc)
[+] Trying pin 12345670
[+] Sending EAPOL START request
[+] Received identity request
[+] Sending identity response
[+] Received M1 message
[+] Sending M2 message
[+] Received M3 message
[+] Sending M4 message
[+] Received M5 message
[+] Sending M6 message
[+] Received M7 message
[+] Sending WSC NACK
[+] Sending WSC NACK
[+] Pin cracked in 3 seconds
[+] WPS PIN: '12345670'
[+] WPA PSK: 
[+] AP SSID:
```

As seen above, hacking WPA/WPA2 is a tedious job in most cases. A word list/dictionary attack may take days, and still, there is a chance it might fail. Also, good dictionaries are huge occupying a lot of memory. An exhaustive brute force may take years depending on the length of the password. Rainbow tables speed things up, but the memory occupied is disastrously large (can be 100s of GBs).